# Comparing KL Divergence and MSE for Covariance Estimation in Target Detection

Daniel Busbib, Tzvi Diskin and Ami Wiesel

2025 IEEE Statistical Signal Processing Workshop (SSP) | 979-8-3315-1800-4/25/$31.00 ©2025 IEEE | DOI: 10.1109/SSP64130.2025.11073417

***Abstract*—Covariance estimation is a core part of adaptive target detection. Most of the works focus on the Mean Squared Error (MSE) metric because it is easy to work with. However, MSE does not always capture the statistical information needed for detection. We advocate for switching to the Kullback-Leibler (KL) divergence. To support this, we analyze the Normalized Signal to Noise Ratio (NSNR) associated with the worst-case target. We show that the KL metric has a structure similar to NSNR and bounds it. To further clarify our point, we derive a simple variant of a classic MSE-based estimator by incorporating KL in a leave-one-out cross-validation (LOOCV) framework. Numerical experiments with various estimators on both synthetic and real data also demonstrate that KL and NSNR behave similarly and are different than MSE. Simply changing the metric in the LOOCV estimator improves KL and NSNR performance while reducing MSE performance.**

## I. Introduction

Covariance estimation is a fundamental problem in statistical signal processing [1]–[6]. A large body of works considers different algorithms in various structures, distributions and settings. This paper addresses the choice of using the Mean Squared Error (MSE) distance vs. the Kullback Leibler (KL) divergence [7] as a performance measure in the context of target detection. The conclusion is that KL, rather than MSE, should be used both in the design and in the analysis of the covariance estimation algorithms.

**Analysis:** Choosing informative metrics is critical for the evaluation of different covariance estimation methods, e.g., [8], [9]. The choice of metric is challenging because covariance estimation is typically an intermediate step within the signal processing pipeline. The estimate is less interesting on its own and is mainly used as an input to a downstream task as target detection, beamforming or Portfolio design [10]. Therefore, accuracy should be measured with respect to the end goal rather than the covariance itself. In the context of statistical signal processing, a key downstream task is detection of a target in additive correlated noise using an adaptive matched filter [11]–[13]. The Signal to Noise Ratio (SNR) controls the probabilities of error. The SNR decreases when an inexact covariance is used. This led to a natural performance metric for covariance estimation known as the Normalized Signal to Noise Ratio (NSNR) [14]–[18]:

$$\mathrm{NSNR} \quad = \frac{\mathrm{SNR}_{\mathrm{estimated}\ \hat{C}}}{\mathrm{SNR}_{\mathrm{true}\ C}} \tag{1}$$

A main difficulty in NSNR is that it depends on the target vector and is only applicable if the target is a priori known. Some works bypass this by assuming that the targets are random [17], [18]. Alternatively, we introduce the worst case NSNR distance with respect to the target vector:

$$d_{\mathrm{NSNR}}(C, \hat{C}) = \max_{\text{target vector}} -\frac{1}{2}\log\ \mathrm{NSNR} \tag{2}$$

and provide a closed form solution to the distance. Interestingly, we show that it has a structure similar to KL and is bounded by it. Therefore, we advocate for using KL as a metric when comparing covariance estimation methods in the context of target detection.

**Design:** Metrics are also important in the design and development of new estimation methods. For example, the seminal LW method and its many extensions are based on data-driven approximations of the MSE which are used to appropriately shrink the covariance matrix [19]–[24]. In addition, an empirical Bayesian shrinkage approach was proposed in [25], improving LW by approximating the Bayes-optimal covariance under a conjugate prior assumption. More recently, in machine learning methods for covariance estimation, metrics are used as loss functions and are part of the training phase of neural networks [26]–[30].

To motivate the use of KL in algorithm design, we develop a variant of LW based on KL instead of MSE. Unlike MSE which is easier to handle, KL involves a challenging matrix inversion. We show that it can be approximated using Leave-One-Out Cross-Validation (LOOCV). Cross validation is a widely used technique for hyper-parameter tuning. By iteratively training on all but one sample and validating on the held-out sample, LOOCV provides an unbiased estimate of unknown KL loss. LOOCV has already been used in MSE-based covariance regularization [24] and we show it can be easily adapted to KL. Following the classical Generalized Cross Validation (GCV) approach [31], we bypass the computationally intensive matrix inversions via the Sherman-Morrison formula. The resulting estimator is a fast and efficient variant of LW focused on minimum KL instaed of MSE.

**Experiments:** We conducted numerical simulations with different covariance estimation methods [19]–[22], [32], [33] in synthetic data as well as in real world hyperspectral data. The goal was to analyze the relations between the MSE, KL and NSNR metrics and the practical detection error probabilities as measured by the Receiver Operating Characteristic (ROC). All the simulations agreed with the theory and showed the resemblance between NSNR, KL and the ROC. This is contrast to the MSE metric which behaved differently. QIS of [32] performed well in all the metrics. LW [19] is

Part of this research was supported by ISF grant 2672/21.

advantageous in terms of MSE but is less recommended in terms of detection performance. Simply changing its objective to KL via LOOCV, reverses this behavior.

## II. Covariance metrics

In this section we discuss and analyze distances between a true positive definite covariance $C$ of size $D \times D$ and its estimate $\hat{C}$. We begin with the popular MSE, then discuss KL and its variants and finally turn to NSNR and its relation to KL.

### A. Mean Squared Error

The most common performance measure parameter estimation is the MSE. In the context of covariances it is usually defined as

$$d_{\mathrm{MSE}}(C, \hat{C}) = \left\| C - \hat{C} \right\|_{\mathrm{Fro}}^2 \tag{3}$$

with the Frobenius norm. This distance is simple, quadratic, differentiable and convex in its arguments. Most of the parameter estimation bounds, as the Cramer Rao Bound, directly address this MSE. On the other hand, it completely ignores basic requirements as positive definiteness of the estimate which are important in practice. Indeed, there are many more realistic variants of the MSE, e.g., the spectral norm [9], [34], or norms on the difference between the inverses of the matrices. For interpretability, it is also common to use a normalized metric denoted by NMSE.

### B. Kullback Leibler divergence

A natural statistical distance is the Gaussian KL divergence [7]:

$$d_{\mathrm{KL}}(C, \hat{C}) = \frac{1}{2}\left(\mathrm{Tr}\left(C\hat{C}^{-1}\right) - D - \log\left|C\hat{C}^{-1}\right|\right) \tag{4}$$

It is easy to see that KL depends on $C$ and $\hat{C}$ via their matrix ratio $Q$ which is the a key component in many covariance estimation metrics (e.g., [33]).

**Definition 1.** *Let $C$ and $\hat{C}$ be two positive definite matrices. Their matrix ratio $Q$ is defined as*

$$Q = \hat{C}^{-\frac{1}{2}} C \hat{C}^{-\frac{1}{2}} \tag{5}$$

*and we denote its eigenvalues by $q_1 \leq \cdots \leq q_D$.*

Using the ratio and the cyclic properties of the trace and determinant operators, KL can be expressed as

$$d_{\mathrm{KL}}(C, \hat{C}) = \frac{1}{2}\mathrm{Tr}\left(Q\right) - \frac{D}{2} - \frac{1}{2}\log|Q| \tag{6}$$

Unlike norms, KL is not symmetric and treats $C$ and $\hat{C}$ differently. Accordingly, some prefer its symmetrized version, also known as Jeffrey's divergence, e.g., [35]

$$d_{\mathrm{J}}(C, \hat{C}) = \frac{1}{2}\mathrm{Tr}\left(Q\right) + \frac{1}{2}\mathrm{Tr}\left(Q^{-1}\right) - D \tag{7}$$

Many downstream tasks, as regression and detection, and their respective algorithms are insensitive to the scaling of the matrices, e.g., NSNR discussed below and Tyler's estimator. This motivates a new variant of the KL loss which is invariant to scaling:

$$\begin{aligned} d_{\text{invariant KL}}(C, \hat{C}) &= \min_{\alpha > 0} d_{\mathrm{KL}}(C, \alpha\hat{C}) \\ &= \frac{D}{2}\log\frac{\frac{1}{D}\mathrm{Tr}(Q)}{|Q|^{\frac{1}{D}}} \\ &= \frac{D}{2}\log\frac{\frac{1}{D}\sum_{i=1}^{D} q_i}{\sqrt[D]{\prod_{i=1}^{D} q_i}} \end{aligned} \tag{8}$$

with the optimal scaling $\alpha = \mathrm{Tr}(Q)/D$.

### C. Normalized Signal to Noise Ratio

Probably the most natural covariance metric in the context of target detection is NSNR (1) from [14]. Specifically, consider the detection of a known signal in additive noise [1], [11]–[14]. The observed vector is

$$x = As + n \tag{9}$$

where $s$ is a length $D$ signal vector and $n$ is a zero mean noise vector with a positive definite covariance matrix $C \succ 0$. The goal is to decide whether $A = 0$ or $A > 0$. The classical adaptive matched filter (AMF) detector for this setting is

$$s^T C^{-1} x \lesseqgtr \gamma \tag{10}$$

where $\gamma$ is a threshold parameter. The test is linear and its performance is characterized by the resulting SNR:

$$\mathrm{SNR}_{\text{true } C} = s^T C^{-1} s \tag{11}$$

In practice, the true covariance matrix of the noise $C$ is rarely known. It is common to replace it with an estimate $\hat{C} \succ 0$. This leads to a mismatched and lower SNR

$$\mathrm{SNR}_{\text{estimated } \hat{C}} = \frac{(s^T\hat{C}^{-1}s)^2}{s^T\hat{C}^{-1}C\hat{C}^{-1}s} \tag{12}$$

To analyze the loss due to the inaccurate covariance, RMB proposed the NSNR [14]:

**Definition 2.** *Let $C$ and $\hat{C}$ be two positive definite matrices. The NSNR in (1) associated with target vector $s$ is given by*

$$\mathrm{NSNR}_s(C, \hat{C}) = \frac{(s^T\hat{C}^{-1}s)^2}{(s^T\hat{C}^{-1}C\hat{C}^{-1}s)(s^TC^{-1}s)}. \tag{13}$$

Ideally, covariance estimation methods try to maximize this NSNR. Note, however, that it depends on $s$ which is not always a priori known.To bypass this dependency, we take a robust approach and consider the NSNR associated with worst possible target.

**Theorem 1.** *Let $C$ and $\hat{C}$ be two positive definite matrices with a ratio $Q$. Then, the worst case NSNR is given by*

$$\min_s \mathrm{NSNR}_s(C, \hat{C}) = \left[\frac{\sqrt{q_1 q_D}}{\frac{q_1 + q_D}{2}}\right]^2. \tag{14}$$

The proof is provided in [36].

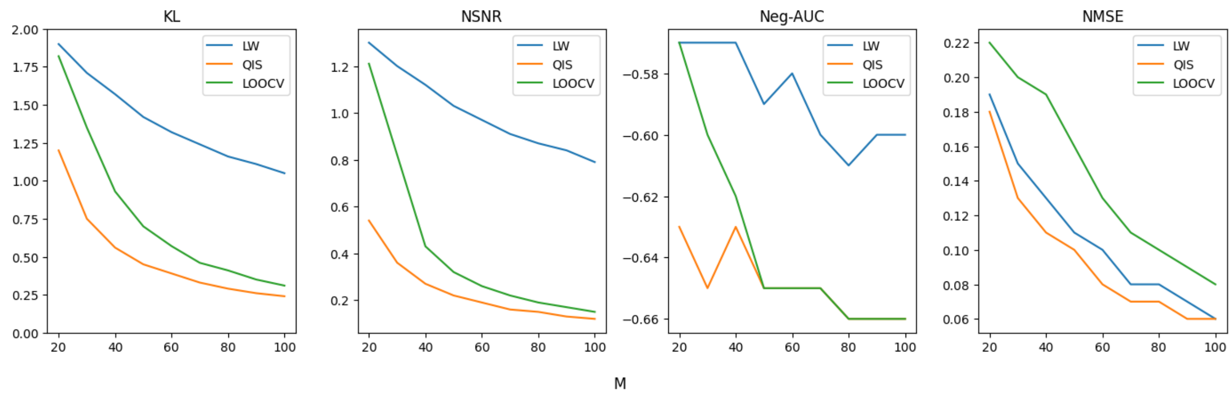


Fig. 1. Synthetic Toeplitz covariance matrices. KL, NSNR and negative AUC metrics are qualitatively similar, and different from NMSE.

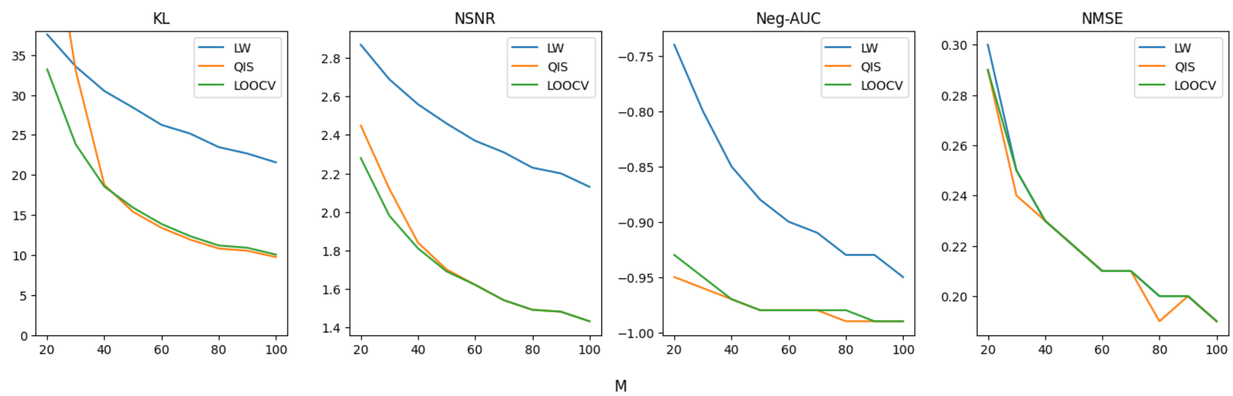


Fig. 2. Real world covariance matrices from the hyperspectral dataset of Pavia University, Trees label. KL, NSNR and negative AUC metrics are qualitatively similar, and different from NMSE.

It is natural to consider the $\text{NSNR}_{\min}$ as a distance function between $C$ and $\hat{C}$. High ratios indicates that $C$ and $\hat{C}$ are close and therefore the metric should be an inversely proportional to the $\text{NSNR}_{\min}$. Following the convention of measuring power ratios in decibels (dB), we also use a logarithm in the NSNR metric.

**Definition 3.** *Let $C$ and $\hat{C}$ be two positive definite matrices. The NSNR distance is defined as*

$$\begin{aligned} d_{\text{NSNR}}(C,\hat{C}) &= -\frac{1}{2}\log\left[\min_s \text{NSNR}_s(C,\hat{C})\right] \\ &= \log\frac{\frac{q_1+q_D}{2}}{\sqrt{q_1 q_D}}. \end{aligned} \tag{15}$$

As expected, $d_{\text{NSNR}} \geq 0$ and is equal to zero if and only if $C = \alpha\hat{C}$ for some $\alpha > 0$. Note the similarity between the NSNR and the KL metrics. All depend on their arguments only through $Q$. The invariant KL and NSNR are both ratios of arithmetic and geometric means of the eigenvalues of $Q$. The only difference is that NSNR uses only the extreme eigenvalues. The next theorem clearly quantifies their relations.

**Theorem 2.** *Let $C$ and $\hat{C}$ be two positive definite matrices. Then*

$$d_{\text{NSNR}}(C,\hat{C}) \leq d_{\text{invariant KL}}(C,\hat{C}) \leq d_{\text{KL}}(C,\hat{C}). \tag{16}$$

The proof is provided in [36].

Therefore, the worst case NSNR seems like a natural metric for covariance estimation in the context of target detection. It is non-differentiable and involves eigenvalues and is therefore less appealing. Instead, KL upper bounds and can be used as a promising surrogate loss.

## III. KL BASED LOOCV SHRINKAGE

Motivated by the previous section, we now derive a simple KL-based covariance estimation method. We emphasize that there are much stronger and more advanced methods addressing a wide range of metrics including KL, e.g., [33]. Our goal is not to compete with these but to demonstrate the importance of choosing the right loss for each task. For this purpose, we simply re-derive the classical LW estimator using an approximate KL loss. The starting point to both estimators is the sample covariance

$$S = \frac{1}{M}\sum_{i=1}^{M} x_i x_i^T \tag{17}$$

Next, both estimators regularize $S$ towards a simpler target matrix

$$\hat{C}_\alpha(S) = (1-\alpha)S + \alpha\frac{\mathrm{Tr}(S)}{D}I \tag{18}$$

LW chooses $\alpha$ by minimizing a data-driven approximation of the unknown MSE. Instead, LOOCV tries to minimize the $d_{\mathrm{KL}}(C, \hat{C}_\alpha(S))$ via leave-one-out-cross-validation (LOOCV). For each $i$, it approximates $C$ as $x_i x_i^T$ and approximates $\hat{C}_\alpha(S)$ as $\hat{C}_\alpha(S_{-i})$ where $S_{-i}$ is the leave-one-out sample covariance

$$S_{-i} = S - \frac{1}{M}x_i x_i^T \tag{19}$$

After omitting constant terms, the problem boils down to

$$\min_{\alpha\in\mathcal{A}} \frac{1}{M}\sum_{i=1}^{M} x_i^T \hat{C}_\alpha^{-1}(S_{-i})\, x_i - \log\left|\hat{C}_\alpha(S)\right| \tag{20}$$

where $\mathcal{A}$ is a finite grid of hyperparameters in $[0,1]$. Computing (20) is expensive due to the many matrix inversions. Fortunately, we can use the Sherman–Morrison formula for efficient rank one updates:

$$\begin{aligned} x_i^T \hat{C}_\alpha^{-1}(S_{-i})\, x_i &\approx x_i^T\left[(1-\alpha)S_{-i} + \alpha\frac{\mathrm{Tr}(S)}{D}I\right]^{-1} x_i \\ &= \frac{z_i}{1 - \frac{1-\alpha}{M}z_i} \end{aligned} \tag{21}$$

where

$$z_i = x_i^T \hat{C}_\alpha^{-1}(S)\, x_i. \tag{22}$$

## IV. Numerical experiments

In this section, we report the results of numerical experiments[1]. We simulated numerous covariance estimation algorithms in different settings and various performance measures. The problem formulation is a standard covariance estimation setting. The input to each algorithm were $M$ samples of a zero mean random vector with an unknown covariance $C$ denoted by $\{x_i\}_{i=1}^M$. The output was the estimate of the covariance $\hat{C}$. We measured performance using four metrics: NSNR in (15), KL in (4), NMSE and area under the curve (AUC). AUC is the most realistic, yet computationally intensive, metric. It is defined as the AUC of the ROC based on thousands of target detection experiments.

We examined many different algorithms including various types of shrinkage and tapering methods. We report the results of the best three[2] which performed well across all the experiments:

- LW [37]: the seminal Ledoit-Wolf shrinkage estimator. It is a weighted average of the sample covariance and the identity where the weight is automatically chosen to minimize a data-driven approximation of the unknown MSE.
- QIS [32]: Quadratic-Inverse Shrinkage estimator. QIS is a more advanced version of LW using non-linear shrinkage of the eigenvalues of the sample covariance.
- LOOCV: the simple KL based variation of LW derived in Section III.

To avoid numerical errors, we projected the outputs of all the estimators to the positive semidefinite cone (with a small $\epsilon = 10^{-2}$ regularization).

### A. Synthetic Data

In the first experiment, we considered the estimation of synthetic Toeplitz covariance matrices. We generated ground truth covariances of size $D = 10$ with random first rows. We then added scaled identity matrices to the covariances to ensure that their minimal eigenvalues were equal to 0.1. For each of these matrices, we generated realizations $x_i \sim \mathcal{N}(0, C)$ for $i = 1, \cdots, M$ and used these as input to the estimation algorithms. The results, averaged over $B = 1000$ independent trials, are provided in Fig. 1. As predicted by Theorem 2, it is easy to see that the KL, NSNR and AUC behaved similarly. QIS was the best algorithm in all the metrics. LOOCV performed worse than QIS but was qualitatively similar. This strengthened our claim that simply changing the objective of LW provided a boost in its NSNR and KL performance. Indeed, LW outperformed LOOCV in terms of NMSE but was significantly worse in terms of NSNR and KL.

### B. Real Data

In the second experiment, we turned to real-world data. We used the hyperspectral dataset from Pavia University [38]. We first divided the dataset into 9 smaller datasets associated with different material labels (Asphalt, Meadows, Gravel, Trees, and others). In each of these, we randomly shuffled the pixels to eliminate spatial effects and obtained a data structure with thousands of vectors $x_i$, which we downsampled to length $D = 35$. We then split the data into two equal subsets. We used the first subset to estimate the ground truth covariance and divided the second subset into many small blocks of $M$ observations $\{x_i\}_{i=1}^M$. Finally, we used these observations to estimate the covariances using the three competing algorithms. The average results were presented in Figure 2. Again, as predicted by the theory, it is easy to see that KL, NSNR and AUC performed similarly.

### C. Conclusions

In all our synthetic experiments (including numerous unreported tests with various structures but always using Gaussian data), QIS was clearly the best algorithm in the KL, NSNR and AUC metrics. However, in many real data experiments with different background materials, QIS often encountered numerical errors at low $M$ and was outperformed by the simpler LOOCV. All the experiments agree with our conjecture that simply changing the loss of LW in LOOCV improves the downstream task performance considerably.

[1] All the experiments and code are provided in open source https://github.com/danielbusbib/KL-Divergence-and-MSE-for-Covariance-Estimation

[2] Another strong estimator was a Random Matrix Theory based algorithm of [33] aimed at directly minimizing the KL distance. It also performed when $M$ was large but was always outperformed by the cheaper QIS and is therefore omitted.

## REFERENCES


[1] S. Kay, *Fundamentals of Statistical Signal Processing: Detection theory*. Prentice-Hall PTR, 1998.

[2] A. Wiesel, T. Zhang *et al.*, "Structured robust covariance estimation," *Foundations and Trends® in Signal Processing*, vol. 8, no. 3, pp. 127–216, 2015.

[3] H. Li, P. Stoica, and J. Li, "Computationally efficient maximum likelihood estimation of structured covariance matrices," *IEEE Transactions on Signal Processing*, vol. 47, no. 5, pp. 1314–1323, 1999.

[4] A. Aubry, A. De Maio, and L. Pallotta, "A geometric approach to covariance matrix estimation and its applications to radar problems," *IEEE Transactions on Signal Processing*, vol. 66, no. 4, pp. 907–922, 2017.

[5] X. Mestre, "On the asymptotic behavior of the sample estimates of eigenvalues and eigenvectors of covariance matrices," *IEEE Transactions on Signal Processing*, vol. 56, no. 11, pp. 5353–5368, 2008.

[6] S. T. Smith, "Covariance, subspace, and intrinsic Cramer-Rao bounds," *IEEE Transactions on Signal Processing*, vol. 53, no. 5, pp. 1610–1630, 2005.

[7] S. Kullback and R. A. Leibler, "On information and sufficiency," *The annals of mathematical statistics*, vol. 22, no. 1, pp. 79–86, 1951.

[8] P. J. Bickel and E. Levina, "Regularized estimation of large covariance matrices," *The Annals of Statistics*, vol. 36, no. 1, pp. 199 – 227, 2008. [Online]. Available: https://doi.org/10.1214/009053607000000758

[9] Y. C. Eldar, J. Li, C. Musco, and C. Musco, "Sample efficient toeplitz covariance estimation," in *Proceedings of the Fourteenth Annual ACM-SIAM Symposium on Discrete Algorithms*. SIAM, 2020, pp. 378–397.

[10] M. Zhang, F. Rubio, D. P. Palomar, and X. Mestre, "Finite-sample linear filter optimization in wireless communications and financial systems," *IEEE transactions on signal processing*, vol. 61, no. 20, pp. 5014–5025, 2013.

[11] E. J. Kelly, "An adaptive detection algorithm," *IEEE transactions on aerospace and electronic systems*, no. 2, pp. 115–127, 1986.

[12] H. Krim and M. Viberg, "Two decades of array signal processing research: the parametric approach," *IEEE signal processing magazine*, vol. 13, no. 4, pp. 67–94, 1996.

[13] D. Manolakis, E. Truslow, M. Pieper, T. Cooley, and M. Brueggeman, "Detection algorithms in hyperspectral imaging systems: An overview of practical algorithms," *IEEE Signal Processing Magazine*, vol. 31, no. 1, pp. 24–33, 2013.

[14] I. S. Reed, J. D. Mallett, and L. E. Brennan, "Rapid convergence rate in adaptive arrays," *IEEE Transactions on Aerospace and Electronic Systems*, no. 6, pp. 853–863, 1974.

[15] D. M. Boroson, "Sample size considerations for adaptive arrays," *IEEE Transactions on Aerospace and Electronic Systems*, no. 4, pp. 446–451, 1980.

[16] B. D. Carlson, "Covariance matrix estimation errors and diagonal loading in adaptive arrays," *IEEE Transactions on Aerospace and Electronic systems*, vol. 24, no. 4, pp. 397–401, 1988.

[17] X. Mestre and M. A. Lagunas, "Finite sample size effect on minimum variance beamformers: Optimum diagonal loading factor for large arrays," *IEEE Transactions on Signal Processing*, vol. 54, no. 1, pp. 69–82, 2005.

[18] O. Besson and F. Vincent, "Performance analysis of beamformers using generalized loading of the covariance matrix in the presence of random steering vector errors," *IEEE Transactions on Signal Processing*, vol. 53, no. 2, pp. 452–459, 2005.

[19] O. Ledoit and M. Wolf, "A well-conditioned estimator for large-dimensional covariance matrices," *Journal of multivariate analysis*, vol. 88, no. 2, pp. 365–411, 2004.

[20] Y. Chen, A. Wiesel, Y. C. Eldar, and A. O. Hero, "Shrinkage algorithms for MMSE covariance estimation," *IEEE transactions on signal processing*, vol. 58, no. 10, pp. 5016–5029, 2010.

[21] E. Ollila and A. Breloy, "Regularized tapered sample covariance matrix," *IEEE Transactions on Signal Processing*, vol. 70, pp. 2306–2320, 2022.

[22] X. Chen, Z. J. Wang, and M. J. McKeown, "Shrinkage-to-tapering estimation of large covariance matrices," *IEEE Transactions on Signal Processing*, vol. 60, no. 11, pp. 5640–5656, 2012.

[23] E. Ollila, D. P. Palomar, and F. Pascal, "Shrinking the eigenvalues of m-estimators of covariance matrix," *IEEE Transactions on Signal Processing*, vol. 69, pp. 256–269, 2020.

[24] L. Xie, Z. He, J. Tong, T. Liu, J. Li, and J. Xi, "Regularized covariance estimation for polarization radar detection in compound gaussian sea clutter," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, p. 1–16, 2022. [Online]. Available: http://dx.doi.org/10.1109/TGRS.2022.3144658

[25] A. Coluccia, "Regularized covariance matrix estimation via empirical bayes," *IEEE Signal Processing Letters*, vol. 22, no. 11, pp. 2127–2131, 2015.

[26] S. Barratt and S. Boyd, "Covariance prediction via convex optimization," *Optimization and Engineering*, vol. 24, no. 3, pp. 2045–2078, 2023.

[27] A. Barthelme and W. Utschick, "DoA estimation using neural network-based covariance matrix reconstruction," *IEEE Signal Processing Letters*, vol. 28, pp. 783–787, 2021.

[28] P. Addabbo, R. Altilio, D. Benvenuti, G. Foglia, and D. Orlando, "A NN-based approach to ICM estimation and adaptive target detection," in *2022 IEEE 12th Sensor Array and Multichannel Signal Processing Workshop (SAM)*. IEEE, 2022, pp. 1–5.

[29] N. Kang, Z. Shang, W. Liu, and X. Huang, "Clutter covariance matrix estimation for radar adaptive detection based on a complex-valued convolutional neural network," *Remote Sensing*, vol. 15, no. 22, p. 5367, 2023.

[30] T. Diskin and A. Wiesel, "Self-supervised learning for covariance estimation," *arXiv preprint arXiv:2403.08662*, 2024.

[31] G. H. Golub, M. Heath, and G. Wahba, "Generalized cross-validation as a method for choosing a good ridge parameter," *Technometrics*, vol. 21, no. 2, pp. 215–223, 1979.

[32] O. Ledoit and M. Wolf, "Quadratic shrinkage for large covariance matrices," *Bernoulli*, vol. 28, no. 3, pp. 1519–1547, August 2022, earlier published as ECON Working Paper No. 335. [Online]. Available: https://doi.org/10.5167/uzh-228054

[33] M. Tiomoko, R. Couillet, F. Bouchard, and G. Ginolhac, "Random matrix improved covariance estimation for a large class of metrics," in *International Conference on Machine Learning*. PMLR, 2019, pp. 6254–6263.

[34] T. T. Cai, C.-H. Zhang, and H. H. Zhou, "Optimal rates of convergence for covariance matrix estimation," *The Annals of Statistics*, vol. 38, no. 4, Aug. 2010. [Online]. Available: http://dx.doi.org/10.1214/09-AOS752

[35] R. Pereira, X. Mestre, and D. Gregoratti, "Consistent estimation of a class of distances between covariance matrices," *IEEE Transactions on Information Theory*, 2024.

[36] T. Diskin and A. Wiesel, "On the normalized signal to noise ratio in covariance estimation," 2024. [Online]. Available: https://arxiv.org/abs/2409.10896

[37] O. Ledoit and M. Wolf, "Honey, i shrunk the sample covariance matrix," *The Journal of Portfolio Management*, vol. 30, 07 2003.

[38] P. Gamba and F. Dell'Acqua, "Pavia university hyperspectral dataset," Online, 2003. [Online]. Available: http://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Pavia_Centre_and_University